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The urbanisation of Java, 2000–2010: towards ‘the island of mega-urban regions’

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ABSTRACT

The mega-urbanisation process in Java is reflected in the spatial patterns of urban population growth between 2000 and 2010, although there has been a small deceleration in the rate of growth recently. This process is also clearly indicated in the significant increase in the number of urban localities, which reflects *in situ* urbanisation and rural–urban transformation in Java. Most districts and cities located adjacent to large cities experienced much higher population growth rates, compared to the core areas in cities. The formation of urban belts with a mix of economic activities connecting large cities is greatly expanding, while the small and medium cities, those with population sizes between 100,000 to one million, have tended to stagnate as their role and functions as centres of socio-economic activities are taken over by the large cities. Java’s mega-urbanisation appears unstoppable, and is largely uncontrolled at the present time. It is a daunting challenge for the central and local governments to manage the spatial urban growth in Java in the near future.

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Introduction

Southeast Asian large cities, including the peripheral areas, have been experiencing tremendous population and physical growth during the last five decades. Urban economic development has spilled over from the urban centres to the fringe areas in a process often referred to as ‘mega-urbanisation’. The built-up areas of urban centres have expanded in all directions beyond the administrative boundaries and include a mix of different economic activities, in which agricultural and non-agricultural activities are often located side by side in places adjacent to urban centres (McGee, 2012; see also Champion & Hugo, 2004). Cities have shifted from mono-centric to poly-centric urban regions.

As Cohen (2004) argued, contemporary urbanisation in the developing world is rather different from what the developed world had experienced in the past, as the former is more strongly affected by the global economy than ever before. As a result of the new globalising economy, the traditional rural–urban distinction is becoming redundant

(p. 23), a few urban centres are experiencing spatial polarisation, and large mega-urban regions are being formed around the centres.

Against the above backdrop, this study examines the macro patterns of urbanisation on the Island of Java, Indonesia – one of the most densely populated areas in Asia–focussing in particular on the phenomenon of mega-urbanisation.

Some analyses of the mega-urbanisation process in Java using the 1990 and 2000 Indonesia's Population Census Data have already been completed (Firman, 1992, 2003, 2004), which enabled this study to update and examine continuity and change in mega-urban formation in Java at the macro level over the period of 2000–2010. In addition, many scholars have also undertaken studies related to rural–urban links and the formation of mega-urban areas in Java utilising the 1980 and 1990 population censuses and population surveys (McGee, 1987, 1991, 2012).

The 2000 and 2010 National Population Censuses showed that the total population of Indonesia increased from 203.5 million to 237.6 million over the period 2000 to 2010, with an average rate of growth of 1.49 per cent per year. Meanwhile, Indonesia's urban population grew from 85.2 million to 118.3 million with an annual growth rate of 3.33 per cent over the same period. About 68 per cent of the urban population lived in Java in 2010, which is only about 7 per cent of the total land area of Indonesia.

Java's urban population grew tremendously from 58.9 million to 80.0 million during the period 2000 to 2010, with an annual rate of growth of 3.10 per cent, as opposed to the total population growth rate of 1.25 per cent per year over this period. Meanwhile, the level of urbanisation, that is, the proportion of urban population, in Java increased significantly from 48.7 per cent to 60.8 per cent over the same period, far above the 49.8 per cent of Indonesia's level of urbanisation (Table 1). In fact, Java is the most densely populated and highly urbanised region in Indonesia (Figure 1). All the Provinces in Java experienced high increase in the level of urbanisation over the period 2000–2010 (Table 1).

This paper is organised into four parts. Part one will critically discuss the process of mega-urbanisation in Asia; Part two will describe the data and methodology used in the study; Part three will analyse the pattern of urbanisation and urban population growth in Java, focussing on the gradual change in localities from being 'rural to urban', urban population growth, shifts in urban employment structure, and development of small towns and medium cities; and Part four will summarise the features of mega-urbanisation in Java and draw a comparison between Java and other developing Asian countries.

Table 1. Percentage of urban population in Java by province, 2000–2010.

No	Province	% Urban 2000	% Urban 2010	Total Population, 2010 (in '000)
1	Jakarta	100.0	100.0	9 607.8
2	West Java	50.3	65.7	43 053.7
3	Central Java	40.4	45.7	32 382.6
4	Yogyakarta	57.7	66.4	3 457.5
5	East Java	40.9	47.6	37 476.7
6	Banten	34.2	67.0	10 632.2
7	Java	48.7	60.8	136 610.5
8	Indonesia	42.2	49.8	237 641.0

Source: Central Board of Statistics² and CBS, 2012.

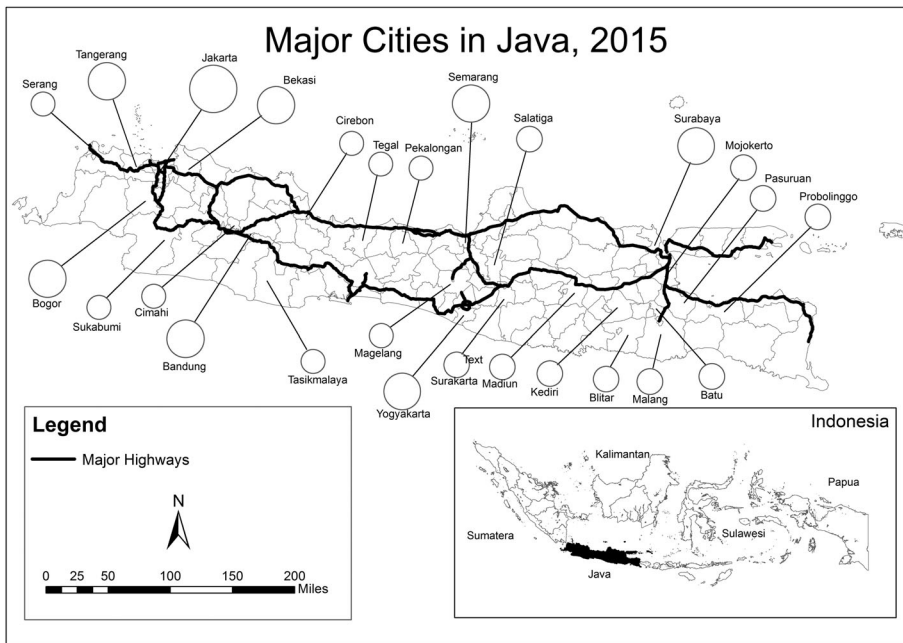


Figure 1. Major cities in Java, 2015.

Asia's mega-urbanisation

The study of Southeast Asian mega-urbanisation was initiated by McGee (1987, 1991) when he coined the word *desakota* from Bahasa Indonesia (Indonesian language) to signify 'rural-urban' mix in the urbanisation process in the region. Peripheral areas of large cities experienced land use conversion from prime agricultural land to urban land uses, a process which has been greatly intensified by the globalisation of the economy, especially in terms of the flows of foreign direct investment (FDI), commodities and people (McGee, 2005).

Mega-urbanisation was characterised by reduced population growth in the urban centres, and increasing population growth in the adjacent areas. According to Brennan-Galvin (2001) this pattern has been identified in many large cities in developing countries, such as Mexico City, Bangkok, Shanghai and Bogota; however, it is not easy to generalise the factors behind the slowing down of population growth in the urban centres (see also Henderson, 2002).

Following the work of McGee (1987, 1991, 2005), many scholars have identified the process of mega-urbanisation in Asia in general, including Webster (1995, 2002), Jones (2002, 2006); Murakami, Zain, Takeuchi, Tsunekawa, and Yokota (2005); Bhagat and Mohanty (2009), and Dupont (2006). There is also research on mega-urbanisation in several large Asian cities such as Beijing and Shanghai (Shen & Wu, 2013; Sit, 2005; Webster, 2001; Xie, Batty, & Zhao, 2007; Yang, 2005); Manila (Ortega, 2012, 2014); Phnom Penh (Percival, 2012); Ho Chi Minh City (Leaf, 2002); Kuala Lumpur (Wong, 2006); and Jakarta (Firman, 2009; Hugo, 2006). It should be noted that the current mega-urbanisation in developing countries is rather different from the urbanisation in

the developed world a century ago as the former has occurred at unprecedented level of urban growth (see McGee, 1995). Urbanisation processes should be understood within the context of current global socio-economic condition. As Brenner and Schmid (2014, p. 20) maintained that ‘the urban is not a universal form but a historical process’.

Data and methods

Studies of urbanisation tend to adopt their own definition of the ‘urban’ (Jones, 2006, p. 247), hence making international comparisons a difficult task (see also Alkema, Jones, & Lai, 2013). Moreover, as Cohen (2004) maintained, the conventional rural–urban dichotomy is inadequate, and the measure of ‘urban’ can potentially become a problem in research of urbanisation (see also UN-Escap & UN-Habitat, 2015, pp. 20–22).

In the Indonesian Population Censuses 2000 and 2010, the smallest administrative unit (*desa*), is considered ‘urban’ based on three criteria: population density; percentage of households engaged in the agricultural sector; and urban facilities and the physical distance to reach them (CBS, 2011a, 2011b; Setiawan, 2009). A technical scoring method has been developed by the Central Board of Statistics to determine if a locality can be categorised ‘urban’ or ‘rural’ on the basis of these three criteria¹. As the definition of ‘urban’ in the Population Census 2000 was similar to that in the 2010 Population Census, it is possible to identify change and continuity in the pattern of urban population growth in Java over the period 2000–2010. However, this approach was unable to identify if the increase in urban population in an administrative unit, such as the *kabupaten* (district), resulted from an increase in the number of *desa-urban* or the actual population.

Some of the problems in using the urban population data in the 2000 and 2010 Indonesia Population Censuses include the fact that the official boundaries of big cities constrained the analysis of the real spread of urban areas (see, Cohen, 2006; Champion & Hugo, 2004; Jones, 2006). It was also likely that there might have been an under-enumeration in the data collection during the Indonesian 2000 census, given the political and economic crises at the time, which caused about 4.5 million people to be excluded from the Census Report (Hull, 2001, 2010). Another problem lay with the definition of urban localities applied in both the 2000 and 2010 Census, which neglected the existence of ‘semi-urban’ localities. These localities have been growing rapidly, especially at the periphery of large cities in Java, and the category ‘semi-urban’ should be added in the next censuses to capture the reality of urbanisation in Indonesia.

The above definition of urban localities was developed from that used in the 1980 and 1990 National Censuses. As Rietveld (1988) argued there were two fundamental problems in the definition of urban areas adopted in the 1980 and 1990 Censuses. First, the criteria of agricultural households was difficult to apply in Java, because in many households the members often had multiple jobs, but for practical reasons, a household was classified as ‘agricultural’ if the primary job of the head of household was in agriculture. Second, the criteria of urban services neglected the quality of the services, which greatly varied from one locality to another (pp. 75–76). These arguments are still relevant to the definition of ‘urban areas’ in the recent Indonesian censuses. Nevertheless, as Hull (2010) argued, Indonesia 2010 Census had a better coverage than the 2000 Census (p. 371), and is the best available data that can be used to analyse urbanisation trends in Indonesia.

The analysis of urbanisation levels in this paper gives attention to the proportion of urban population, urban population growth, and urban employment structure in all municipality cities in Java and the adjacent areas. Using data from the 2000 and 2010 National Census, the paper examines the spatial pattern of urbanisation in Java by focussing on the gradual changes in localities from being 'rural' to 'urban', the continuity of mega-urbanisation, shifts in urban employment structure, and the development of small towns and medium cities.

The patterns of urbanisation in Java

In-situ urbanisation: the gradual change of localities from being 'rural' to 'urban'

The urban localities in Java increased from 2533 to 3641 over the period 1980–1986 (Central Board of Statistics, 1988), and the number continued to increase from 7510 to 9239 over the period 2000–2010. In 2000, the number of urban localities was about 30 per cent of the total localities in Java, whereas in 2010, the number increased to nearly 37 per cent, indicating how fast urban localities had grown and also reflecting the *in situ* urbanisation process in Java, resulting from gradual changes from rural to urban localities. In comparison, for Indonesia as a whole the figure was only 18.0 per cent and 20.5 per cent in 2000 and 2010, respectively. The gradual change of localities from being 'rural' to 'urban' essentially reflects rural–urban transformation in Java in general. A similar phenomenon of in-situ urbanisation has also been taking place in rural China, where rural centres are growing to become small urban centres, as a result of mostly the development of small industries and other non-agricultural activities in the rural villages (see Zhu, 2000a, 2000b, 2004).

While urban growth has been very rapid, urbanisation in Java is considered a recent phenomenon, as Christie (1991) argues:

... despite its heavy population, Java has not until recently been significantly urbanized. The 1815 figures indicate that at the time there were only five towns on the island with more than 20,000 inhabitants; at the end of the nineteenth century, towns at that size or larger still contained less than 3 percent of the population, and 90 percent of the population lived in communities of fewer than 5,000 inhabitants [Boomgaard, 1989, p. 111], despite the administrative and economic restructuring and rapid growth of rail links brought by the Dutch colonial government (pp. 23–24).

For comparison, the level of urbanisation in Java in 1961 was 15.6 per cent (see Nitisastro, 2006, p. 178) whereas in 1980 and 1990, the level rose to 25.1 per cent and 35.7 per cent respectively (Firman, Kombaitan, & Pradono, 2007). This further increased to 48.7 per cent and 60.8 per cent in 2000 and 2010, respectively (Firman, 2016). These figures essentially reveal that urbanisation in Java has been growing even faster since the end of the twentieth century.

Continuity in mega-urbanisation

By 2010, there were 12 cities which had a population size of one million and more in Indonesia. Nine of these cities are located in Java, including Jakarta, Surabaya, Semarang, Bandung, Tangerang, South Tangerang, Bekasi, Bogor and Depok. Apart from Surabaya,

Bandung and Semarang, the other six are located within the Jakarta Metropolitan Area (JMA) (Figure 1). This resulted in about 20 per cent of Indonesia's population living in JMA in 2010, which constituted about 31 per cent of Java's urban population. In other words, both Indonesia's and Java's urban population is highly concentrated in the JMA, which makes this metropolitan region a primate city. The rate of population growth of the JMA reached 3.60 per cent per annum over the period 2000–2010, much higher than the 3.10 per cent per year for Java's urban population over the same period.

The annual population growth rate of these cities with population sizes of one million or more is higher than Java's average annual rate of 1.25 per cent, with the exception of Bandung City (1.19%). Depok (4.25%), Bekasi (3.44%), Tangerang (3.20%), and Jakarta (1.40%) all of which are located in the JMA, and Semarang (1.44%) are all above the average (Table 2). The United Nations had predicted that Jakarta City's annual rate of population growth could reach 3 per cent, and the total population 17.3 million by 2015 (see Brennan-Galvin, 2001, p. 19), but results from the 2010 Census indicates that this prediction seems to be largely overestimated.

The relatively low rate of population growth of Jakarta, Surabaya and Bandung (Figure 1) was basically the result of the fact that socio-economic development led to faster population growth in the outskirts of the core cities. For example, the rate of annual population growth in Jakarta City, the core of the JMA, reached only 1.40 per cent over the period 2000–2010, whereas the rate of annual population growth in the peripheral areas was much higher, including in Bekasi District (4.70%), Depok City (4.33%), Tangerang City (3.50%), Bekasi City (3.44%), Bogor City (3.15%), and Bogor District (2.40%) (Figure 2, see also Table 2).

In contrast, the annual rate of population growth in Jakarta City was 5.5 per cent over the period 1930–1961 (Milone, 1964), 4.4 per cent during 1961–1971, and 4.0 per cent during 1971–1980 (Hugo, 1982, p. 63). The ratio of core city population to the population of the total metropolitan area has also decreased substantially, as seen in the decline of the ratio of Jakarta City population to JMA's population from 55 per cent in 1990 to 43 per cent in 2000 (Firman et al., 2007), followed by a further decline to 36 per cent in 2010. Meanwhile, the population density of the JMA had increased from 37.6 persons per square kilometre in 2000 to 44.6 in 2010, while the population density of Jakarta City increased even more tremendously from 128.0 in 2000 to 145.9 in 2010 (Salim, 2013).

Table 2. Population and annual population growth rate of Java's large cities, 2000–2010.

City (Figure 1)	Population ('000)		Annual Growth Rate (%)
	2000	2010	
Jakarta (*)	8356.5	9604.3	1.49
Surabaya	2578.1	2715.5	0.53
Bandung	2140.0	2394.8	1.19
Bekasi (*)	1663.8	2334.8	4.03
Semarang	1342.6	1535.9	1.44
Tangerang (*)	1311.8	1798.6	3.71
Depok(*)	1146.0	1738.6	5.17
South Tangerang(*)	n.a.	1290.3	n.a.
Bogor(*)	743.5	950.3	3.20

(*) Located in Jakarta Metropolitan Area (Jabodetabek).

Source: Central Board of Statistics, 2011b³ and Firman, 2004.

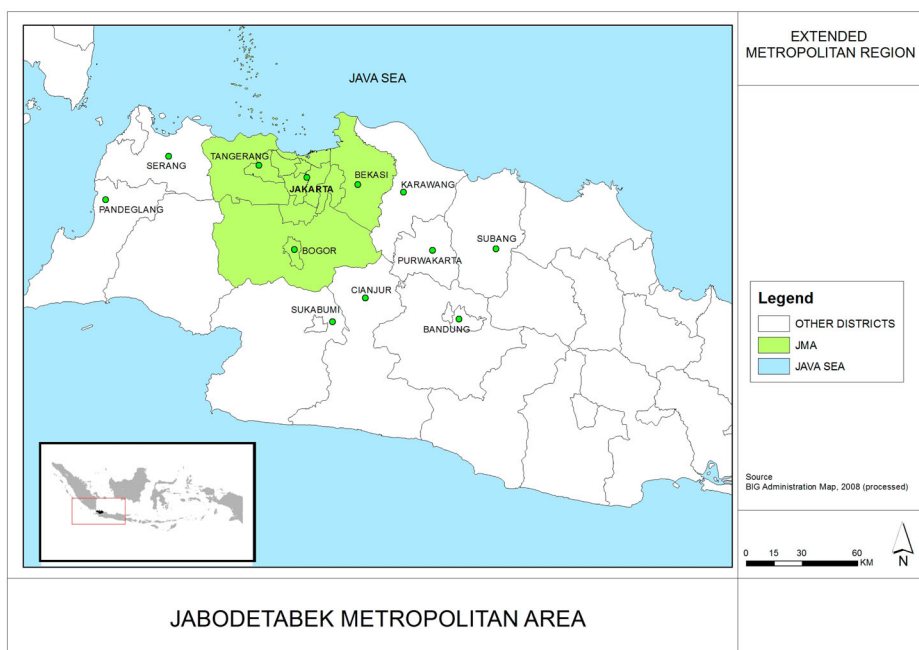


Figure 2. Jakarta Metropolitan Area.

For Jakarta City, in-migration, as indicated by place of residence at the time of census enumeration being different from place of birth, was 644,000 in 2010, in contrast to the recent out-migration reaching approximately 883,500 (Statistical Yearbook of Indonesia, 2012; cf. Firman, 2016). In other words, Jakarta City experienced negative lifetime migration over the period of 2000–2010 as migrants began to be diverted from Jakarta City to the surrounding areas as their destinations. This situation had occurred since the 1980s, when Jakarta City began to experience a turnaround from a net-positive to a net-negative five year migration rate between 1980 and 2000 (Firman, 2004, p. 428; see also Firman, 1994).

The spatial pattern of urban population growth in the Bandung Metropolitan Area (BMA) also showed a similar pattern to that of the JMA. The annual rate of population growth of the City of Bandung, the core of the BMA, was only 1.11 per cent over the period 2000–2010, while the surrounding areas grew much faster, including Cimahi City (2.06%), Bandung District and West Bandung District (1.29%) during the period. For comparison, the annual population rate of growth of Bandung City reached 5.9 per cent during the period 1930–1961 (Milone, 1964, p. 1005; see also Colombijn, 2010, p. 66 & Appendix 1). Just like the JMA, the proportion of the population of Bandung City to the total population of the metropolitan area declined from 34 per cent in 1990 to 29 per cent in 2000, and went down further to only 28 per cent in 2010 (Firman, 2016).

The recent physical development of both the BMA and the JMA (Figure 1) was marked by the formation of an urban belt from Bandung to Jakarta City of about 200 kilometre long, characterised by a mixture of socio-economic activities, including industrial, residential, and agricultural activities, which blurred the rural–urban distinction in these areas

(Firman, 2008, 2009; Firman & Dharmapatni, 1995). The population growth was evident in the development of the Bandung-Jakarta urban corridor, and commuting also played an important role where nearly 3 million people commute daily between Jakarta City and the adjacent areas by trains, personal cars and buses in the early 2000s (Hidayat, 2007).

A study employing remote sensing and geographic information system (GIS) techniques showed that the built-up areas of Jakarta City increased from 560 to 594 square kilometres during 2000–2010, whereas those in the peripheral areas of the JMA, including in Bogor, Tangerang and Bekasi Districts increased much more significantly from 544 to 850 square kilometres over the same period (Salim, 2013). Meanwhile the built-up areas in the BMA increased substantially from 342.8 square kilometres in 1992 to 658.1 square kilometres in 2012 (Ardiwijaya, Soemardi, Suganda, & Temenggung, 2014, p. 209). This reflects how fast the urban physical development in the fringe areas of Jakarta and Bandung Metropolitan Areas is growing.

The Gross Regional Domestic Product (GRDP) of the JMA contributed about 26 per cent to the national Gross Domestic Product (GDP) per year over the period 1990 to 2006 (Table 3), whereas Jakarta City's share to the JMA's GDP was 71 per cent in 2008, or Rp. 350.6 trillion (World Bank, 2011a, 2011b). In contrast, the Surabaya Metropolitan Area, the second largest metropolitan area in Indonesia, only contributed about 7 percent on average per year over the same period, while the Bandung Metropolitan Area and the Semarang Metropolitan Area contributed even smaller figures of 1.9 per cent and 1.8 per cent per year respectively. These figures showed the importance of mega-urban regions in Java, especially the JMA and, to a lesser extent, the Surabaya Metropolitan Area, to the national economy.

The annual population growth of Surabaya City, the core of Surabaya Metropolitan Area (Gerbangkertasusila) was only 0.53 per cent over the period 2000–2010 (Table 2 and Figure 1), compared to 3.5 per cent during 1931–1961 (Milone, 1964, p. 1005). Meanwhile, the fringes of this metropolitan area experienced high annual population growth, such as Gresik District (1.59%), Mojokerto District (1.20%), Sidoarjo District (2.21%), and Bangkalan District (1.20%) over the 2000–2010 period. Besides the development of the Surabaya Metropolitan Area, the fast development of the Surabaya-Malang (Figure 1) urban corridor also characterises recent urban transformation in East Java.

The population of Semarang City grew by 1.44 per cent over the period 2000–2010 (Table 2), compared to 2.77 per cent during 1930–1961 (Milone, 1964, p. 1005). While Semarang City as the core of the Semarang Metropolitan Area (Figure 1) experienced a moderate population growth, the fringe areas of the city also had a low population growth rate over the period 2000–2010, including Districts of Kendal (0.66%) and Demak (0.88%) and Salatiga City (1.30%). This indicates that the Semarang Metropolitan

Table 3. Percent of national GDP by Metropolitan areas in Java, 1993–2006.

Agglomeration (Figure 1)	Population 2005 (million)	1999 (% of the National GDP)	2003 (% of the National GDP)	2006 (% of the National GDP)
Jakarta (Jabodetabek)	25.80	26.1	26.3	25.5
Surabaya (Gerbangkertosusila)	10.36	6.7	6.8	7.1
Bandung (Bandung Raya)	6.98	1.5	2.0	2.2
Semarang (Kedungsepur)	5.01	1.5	2.0	1.9

Source: The National Planning Agency (Bappenas); the World Bank Jakarta Office, 2011a, 2011b.

Area did not experience mega-urbanisation to the same extent as Jakarta, Bandung and Surabaya. Nevertheless, Handayani and Rudiarto (2014) noted that some peripheral areas of the Semarang Metropolitan Area, notably those located from 17 to 20 kilometres from Semarang City experienced an increasingly higher population growth rate, while the Remote Sensing and GIS analysis showed that it experienced rapid development of new urban settlements in the peri-urban areas (p. 85).

Yogyakarta City, a medium-sized city and the capital of the Province of Yogyakarta Special Region (Figure 1), had a negative annual population rate of growth, of 0.23 per cent over the period 2000–2010. In other words, the city experienced population loss while the fringes of the city, that is, Kabupaten Sleman and Kabupaten Bantul, had an annual population growth of 1.56 per cent and 1.37 per cent respectively over the same period (Giyarsih & Alfana, 2013).

The growth of the urban population in the periphery of large cities in Java need not be the result of migration from the city centre to the outskirts (see Browder, Bohlan, & Scarpaci, 1995), but could also result from the reclassification of localities previously defined 'rural' to become 'urban' (Gardiner & Gardiner, 2006; Salim & Firman, 2011). Jones, Tsay, and Bajracharya (1999) specifically called this phenomenon 'Metropolitan Turnaround' (p. 5), in which the population growth in the core city area experience deceleration, while the population in the outskirts grows much faster. A study of urban population growth in Java in the 1980–1990s showed that although it was not entirely clear, the urban population growth during that period created an urban belt along the north coast of Java which stretched from Jakarta to Surabaya through Cirebon and Semarang. A similar process also took place in creating the belts of Jakarta-Bogor, Surabaya-Malang and Yogyakarta-Semarang (Firman, 1992, p. 104; Figure 1). The study also indicated the disparity in urban growth rates between the north and south coasts of Java, as Kabupaten (Districts) with high increase in the proportion of urban population were located on the north coast, while only a few were found on the south coast. As Hall and Pfeiffer (2000) argued the massive urban belt in the north coast of Java consists of scores of highly networked cities of different sizes (p. 5).

The spatial pattern of the urban population that developed in Java from 1990–2000 revealed a continuity of mega-urbanisation trends since the 1980s (Firman, 2003). However, the growth in the urban population over the period was slightly lower than in the 1980–1990 period, except for cities which experienced administrative boundary expansion. The study also noted that most Kabupaten located adjacent to large cities, notably in Jakarta, Surabaya, Bandung and Yogyakarta metropolitan areas (Figure 1), experienced high population growth. Therefore, it would be misleading to consider the annual rate of growth of large cities in Java without taking into consideration the population growth in the Kabupaten in the surrounding areas (p. 63). The study revealed that continuity in mega-urban development during the period 1990–2000 was also characterised by the deceleration of urban population growth.

In Java, the process of mega-urbanisation has been present for almost three decades ago (Firman, 1992, 2003, 2009; McGee, 1987, 1995, 2012; McGee & Robinson, 1995). Nonetheless, the annual urban population growth in Java has decreased from 4.38 per cent over the period 1990–2000 to 3.17 per cent during the period 2000–2010, but it should be kept in mind that the deceleration of urban population growth in Java could be due to the decline of the total fertility rate in Indonesia from 5.7 in 1967–1969 (Adioetomo, 1997)

to 2.6 in 2002 (Population Reference Bureau, 2002), and to 2.3 in 2012 (Population Reference Bureau, 2012).

Shifts in urban employment structure

The structure of employment, calculated based on the population of 15 years of age and more who have a job during the census enumeration time (CBS, 2011b), showed that the core cities of mega-urban regions were largely dominated by the sectors of trade and service, while manufacturing only played a relatively insignificant role (Table 4). Jakarta City had employment records of 4,309,000 in 2010, consisting of Manufacturing (15.6%); Trade (25.4%); and Services (36.9%); and Other Sectors, including Agriculture, Forestry and Extractive (22.1%), whereas the employment figures in Surabaya city reached 1,241,600, consisting of employment in Manufacturing (16.6%); Trade (28.5%); Services (42.3%); and Other Sectors (12.6%). Likewise the employment structures in Bandung City and Semarang City basically displayed similar patterns.

In 1980, almost half of the employments in medium and large-scale industries, such as those employing more than 20 workers, in Java were located in the cities, and one-quarter in the Kabupaten (District) adjacent to the large cities (Jones, 1984, p. 143). This indicated the primary role of large cities as centres of manufacturing activities at the time. In more recent times, the change in employment structure in Java's large cities reflected the shifting of the core cities in large metropolitan areas from centres of manufacturing activities to become centres of trade and service activities. This included trade and services in the informal sectors, as the manufacturing activities have largely moved to the fringes and other areas (see Hudalah & Firman, 2012; Hudalah, Viantari, Firman, & Woltjer, 2013) (Figure 1).

Small town and medium cities

Small towns and medium cities play an important role in urbanisation and regional development in developing countries (Roberts, 2014; Satterthwaite & Tacoli, 2003; Tacoli, 2003; Veron, 2010), as discussed in the work of Watts (1963) and Rondinelli (1983a, 1983b, 1983c, 1986). The question to be addressed in this study is the extent to which small towns and medium cities play a role in Java's mega-urbanisation.

The annual rate of population growth of small and medium cities in Java, arbitrarily defined as those with population size between 100,000 to one million, is relatively low, below Java's total population growth of 1.34 per cent per year over 2000–2010 (Table 5), with the exception of Serang and Cilegon in the Province of Banten which could actually be considered as part of the JMA because of proximity and high accessibility to Jakarta City; Cimahi, which is part of Metropolitan Bandung; and Sukabumi in the

Table 4. Employment structure of large cities in Java, 2010.

City (Figure 1)	Total Employment ('000)	% of Manufacturing Sector	% of Trade Sector	% of Service Sector
Jakarta	4 309.0	15.6	25.4	36.9
Surabaya	1 241.6	16.6	28.5	42.3
Bandung	952.8	18.7	28.6	33.2
Semarang	699.7	17.6	24.9	47.4

Source: Calculated by the author based on 2010 Population Census Data (Central Board of Statistics, 2011b).

Table 5. Population and annual population growth rate of municipality small towns and medium cities in Java by Province, 2010.

City or Town (Figure 1)	Province	Population	Rate of Growth 2000–2010 (%)
Sukabumi	West Java	299,247	1.72
Cirebon	West Java	295,764	0.84
Cimahi	West Java	541,139	2.06
Tasikmalaya	West Java	634,424	1.11
Banjar	West Java	175,165	1.14
Magelang	Central Java	115,094	0.74
Surakarta	Central Java	496,993	0.25
Salatiga	Central Java	169,348	1.13
Pekalongan	Central Java	280,878	0.63
Kediri	East Java	257,892	0.90
Blitar	East Java	131,052	1.01
Malang	East Java	812,983	0.80
Probolinggo	East Java	216,001	1.26
Pasuruan	East Java	184,690	1.02
Mojokerto	East Java	119,781	0.49
Madiun	East Java	169,959	0.56
Batu	East Java	190,155	1.22
Cilegon	Banten	374,559	2.44
Serang	Banten	557,785	2.88
Yogyakarta	Yogyakarta	388,627	–0.23

Source: Calculated by the author based on 2010 Population Census Data (Central Board of Statistics, Result of 2010 Population Census).

Province of West Java. It should be noted, however, that cities presented in Table 5 are limited to those which possess the status of ‘Municipality’ (Kota Otonom), as the Indonesian Census do not take into account the population of ‘Non-Municipality’ Cities even if the population exceeds 100,000. Some examples of the latter category are the City of Garut, Ciamis and Cianjur in West Java; Purwokerto and Klaten in Central Java; Sleman and Bantul in Yogyakarta; Sidoarjo, Gresik and Bangkalan in East Java.

The small towns and medium cities in Java play a less significant role as secondary centres of economic activities, such as the distribution and collection of goods (Hinderink & Titus, 1988, 2002; Titus, 1993), as these functions have been taken over by larger cities because of the proximity between large, medium and small towns and cities in Java, (usually less than 100 kilometres). In fact, farmers and peasants often deal directly with tradesmen, wholesalers and suppliers from large cities, bypassing rural centres and small towns (Van der Wouden, 1997). According to the World Bank (2011b), Indonesia’s medium cities with population between 100,000 and 500,000, including those in Java, have the least agglomeration economies than cities of any other size (p. xiii). The medium cities are hence experiencing negative economic growth and declining productivity, while the small towns in Java are often considered ‘Pension Towns’ where residents comprise many elderly or retired persons (Titus, 1993; Van der Wouden, 1997; see also Firman et al., 2007). As Gardiner (1997) argued more than 25 years ago, while large cities in Java grew rapidly, the small towns and medium cities tended to stagnate.

Concluding observations

Mega-urbanisation is a recent phenomenon in Java; nonetheless it is growing very rapidly. Within this context, the study has analysed the spatial patterns of urbanisation in Java over

the period 2000–2010. More recently, this process is taking place with slower urban population growth, as the annual urban population growth in Java has declined from 4.38 per cent over the period 1990–2000 to 3.17 per cent in the period 2000–2010. However, it should be noted that the deceleration of urban population growth in Java may be attributed to the decline in the total fertility rate in Indonesia from 5.7 in the mid 1960s to 2.5 by the end of 2000s.

Mega-urbanisation in Java is largely characterised by an increasing number of ‘urban’ localities, from about 30 per cent of the total localities in 2000 to 37 per cent in 2010, as compared to the increase in Indonesia as a whole from 18 per cent and 20.5 per cent over the same period. This clearly reflects the importance of *in situ* urbanisation in the context of rural–urban transformation in Java.

The function of large cities, including Jakarta, Bandung, and Surabaya, which represent the core of metropolitan areas, has shifted from manufacturing in the past to trade and service activities, as manufacturing activities moved to the fringe areas. In fact, urban areas has greatly expanded and radiated far beyond administrative boundaries in almost all directions from the core cities. Most of the Kabupaten located adjacent to large cities, notably in Jakarta, Surabaya, Bandung and Yogyakarta Metropolitan Areas, experienced high population growth, with the exception of Semarang Metropolitan Area. Therefore, it would be misleading to consider the annual growth rate of large cities in Java without taking into consideration the population growth of the Kabupaten in the surrounding areas.

Mega-urbanisation in Java is physically reflected in the intensified formation of several massive urban belts, connecting large cities, most notably, Jakarta-Bandung; Jakarta-Cirebon-Semarang-Surabaya; Surabaya-Malang; and Yogyakarta-Semarang, in creating ‘mega-urban regions’ (see McGee & Robinson, 1995). This phenomenon is exemplified in the formation of an urban belt of 200 kilometres connecting the JMA to the BMA. It resulted in a mixture of socio-economic activities, and blurred the rural–urban distinction in these areas. Population growth is evident from the growing volume of commuters in this massive urban corridor. This urban belt also plays a very important role in the national economy.

The recent mega-urbanisation processes in Java were also characterised by the stagnation of small towns and medium-sized cities as secondary centres of economic activities, especially for the distribution and collection of goods. Most likely their role in these functions have been taken over by the larger cities because of good accessibility and close physical distances between large and small towns and medium cities in Java.

In sum, the mega-urbanisation trends in Java are similar to those occurring elsewhere in Asia. Nevertheless, there are several specific distinctive features, which may not be shared by other Asian countries: (1) Mega-urban regions in Java continue to grow not only in the largest urban agglomeration Jabodetabek Metropolitan, which is the capital city, but also in the secondary cities, including Surabaya and Bandung Metropolitan albeit on a smaller scale; (2) In-situ urbanisation is taking place very rapidly, as indicated in the changes of rural localities to become urban localities, which reflects rural–urban transformation in Java; (3) Metropolitan areas are connected by urban belts, particularly in the Jakarta-Bandung Mega-Urban Regions; (4) Employment structure in the urban centres is mainly dominated by the service sector, while the fringe areas are dominated by manufacturing and industrial sectors; and (5) Medium cities and small towns, as represented by those which have municipality (*kota*) status, are growing very slowly.

Java's mega-urbanisation appears unstoppable, and seems to be an inevitable consequence of socio-economic development. From the perspective of urban planning, the problem lies in that it is largely uncontrolled and unsustainable, which could result in serious environmental problems in the near future. These include haphazard urban growth, the conversion of prime agricultural land to other uses, traffic congestions, air and water pollution, and flooding, if no policy action is taken to deal with the problems. It is therefore a daunting challenge for the central and local governments to manage mega-urban growth and physical development in Java at present and in the future.

Notes

1. The population density score is between 1 for population density less than 500 people per square kilometre, to 8 for over 8500 persons per square kilometre. The score for households engaged in agricultural sectors ranges from 1 for percentages over 70 per cent, to 8 for those less than 5 per cent, whereas the score for access to urban facilities is either 0 or 1 kilometre. Based on this scoring method, a locality which has a total score of 10 and over is considered an urban locality (*desa urban*) or a rural locality (*desa perdesaan*)?
2. See, Central Board of Statistics, <http://www.bps.go.id>
3. See, Central Board of Statistics (2011), <http://www.bps.go.id>.

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